

IBOutlet Exercises

First Challenge - Lamp

Create an application that simulates being the control of a lamp.

Upon entering the application, a launch screen should be displayed. Once the application is open a black screen should be displayed to simulate that the lamp is turned off.

The interface should include a button that toggles between *Turn On* and *Turn Off* according to the current state of the lamp.

If the button displays *Turn On*, pressing it should change the background of the screen to white, simulating that the lamp has been turned on. If the button displays *Turn Off*, pressing it should change the background of the screen to black, simulating that the lamp has been turned off.

Extra Challenge (optional)

To enhance the user experience, two images related to a lamp states should be included. These images should be visible according to the lamp's state (*on* or *off*).

Note: Ensure that you implement the logic for changing the background and the visibility of the lamp images consistently with the current state of the lamp.

Second Challenge - RGB Colors

Develop an application that allows users to explore and adjust RGB colors using sliders.

The application will consist of three sliders, each representing one of the RGB color components (Red, Green, and Blue).

- When any of the sliders is moved, the application should update the background color in real-time. This will allow users to experiment with different color combinations interactively.
- Next to each slider, a label will be displayed indicating the numerical value selected on that slider.

Add a "Reset" button that will allow users to revert to the default color values (for example, black).

Third Challenge - Login

In the [login project from the storyboard-exercises \(https://github.com/iOSLabUNAM/storyboard-exercises#login\)](https://github.com/iOSLabUNAM/storyboard-exercises#login), is necessary to connect all components and provide functionality.

A users should be able to access using the following login information:

Username: Diplomado2024
Password: YourName_BirthMonth

Consider the following points:

- If the text fields are empty, the *Login* button should be disabled.
- When data is entered in both fields, the button should be automatically activated.
- In case the username or password is incorrect, a native alert should be displayed informing the user that the data is incorrect.
- If the login information is correct, the user should be redirected to a welcome screen, where a welcome message should be displayed along with their username.

Extra Challenge (optional)

The password text field should have the ability to hide the entered text. This should be implemented using the native password hiding feature to enhance application security.

Note: *If you haven't completed this exercise in the previous attempt, we encourage you to finish it and continue working on it to achieve a successful implementation*